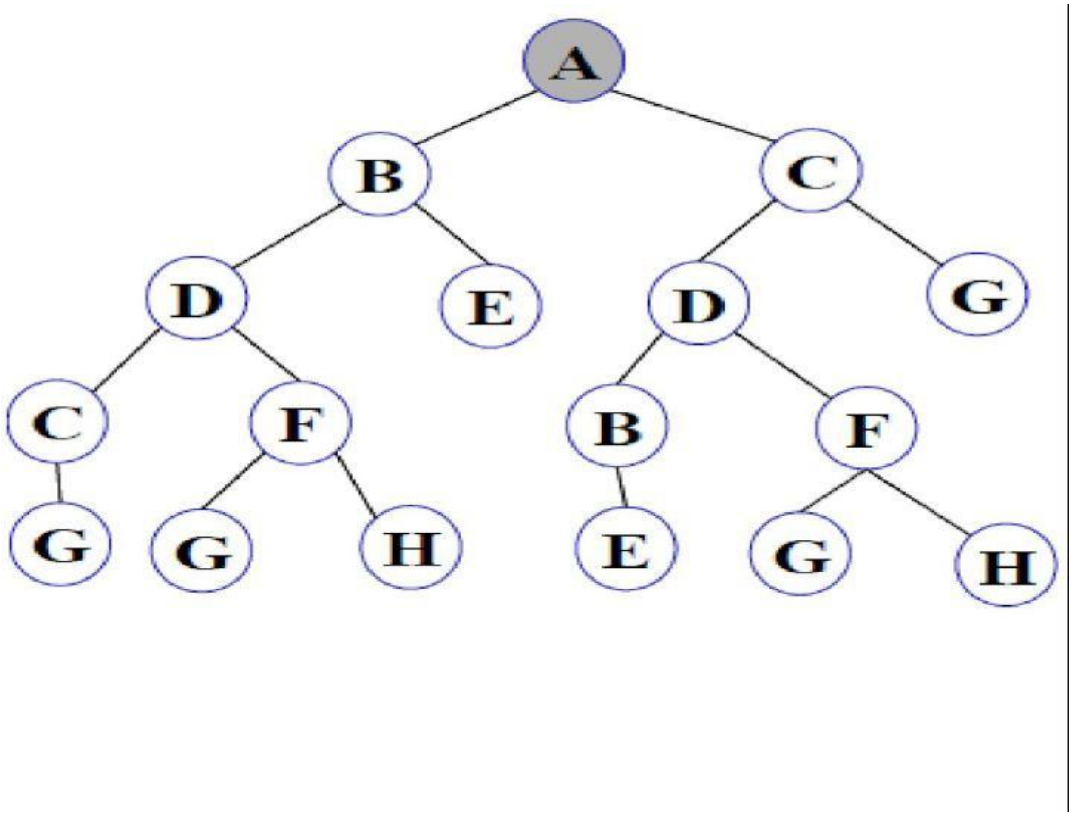
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Task: Breadth-First Search (Bfs) And Depth-First Search (Dfs)

BREADTH-FIRST SEARCH (BFS) AND DEPTH-FIRST SEARCH (DFS) ON A GIVEN STATE SPACE TREE



1. Introduction

Breadth-First Search (BFS) and Depth-First Search (DFS) are two fundamental uninformed search algorithms used in Artificial Intelligence to explore a state space and find a required node.

The given state space tree has A as the root node. The tree contains several nodes labelled B, C, D, E, F, G, and H. Some labels appear more than once because they represent different nodes at different positions in the tree.

The objective is to demonstrate how BFS and DFS explore the given tree and determine their traversal orders.

Assumption: The nodes are explored from left to right, as shown in the diagram.

2. Breadth-First Search (BFS)

Breadth-First Search explores the tree level by level. It visits the root node first, then all nodes at the next level, followed by the nodes at the subsequent levels.

BFS uses a queue (FIFO — First In, First Out) to store the nodes waiting to be explored.

Steps involved in BFS

1. **Start at the root node A.**
2. **Visit A and add its children B and C to the queue.**
3. **Remove B from the queue and visit its children D and E.**
4. **Remove C from the queue and visit its children D and G.**
5. **Continue processing the nodes in the queue, visiting each node's children from left to right.**
6. **Stop when all reachable nodes have been visited.**

BFS traversal of the given tree

The nodes are visited level by level as follows:

- **Level 0: A**
- **Level 1: B, C**
- **Level 2: D, E, D, G**

- **Level 3: C, F, B, F**
- **Level 4: G, G, H, E, G, H**

Therefore, the BFS traversal order is:

A → B → C → D → E → D → G → C → F → B → F → G → G → H → E → G → H

Explanation

BFS first visits A, the root node. It then visits B and C, which are the children of A. Next, it visits the children of B and C from left to right. The same process continues for the lower levels until every node in the tree has been visited.

BFS is useful when the objective is to find the shallowest goal node in an unweighted search tree.

3. Depth-First Search (DFS)

Depth-First Search explores one branch of the tree as deeply as possible before returning to explore another branch.

DFS can be implemented using a stack (LIFO — Last In, First Out) or recursion.

Steps involved in DFS

1. **Start at the root node A.**
2. **Visit A and move to its left child B.**
3. **Continue down the leftmost branch until a leaf node is reached.**
4. **Backtrack to the nearest node with an unexplored child.**
5. **Explore that child and continue searching deeply.**
6. **Repeat the process until all nodes have been visited.**

DFS traversal of the given tree

Following the left-to-right order, the nodes are visited as follows:

1. **Start at A.**
2. **Move to B.**
3. **Move to the left child D.**
4. **Move to C.**
5. **Move to G, then backtrack.**
6. **Return to D and explore F.**
7. **Visit G, then H.**
8. **Backtrack to B and visit E.**
9. **Return to A and move to C.**
10. **Visit its left child D, then B and E.**
11. **Backtrack to D and explore F.**
12. **Visit G, then H.**
13. **Finally, visit G, the right child of the upper C.**

Therefore, the DFS traversal order is:

A → B → D → C → G → F → G → H → E → C → D → B → E → F → G → H → G

Explanation

DFS begins at A and explores the left subtree completely before moving to the right subtree. Whenever it reaches a leaf node, it backtracks to the nearest ancestor that has an unexplored child. This continues until all branches have been explored.

DFS is useful for exploring deep paths and can require less memory than BFS when the search tree is very wide.

4. Difference Between BFS and DFS

Feature	Breadth-First Search (BFS)	Depth-First Search (DFS)
Search method	Level by level	Branch by branch
Data structure	Queue (FIFO)	Stack (LIFO) or recursion
Starting node	A	A
Exploration order	Visits all nodes at one level before going deeper	Explores one branch deeply before backtracking
Completeness	Complete for finite branching trees	Complete for finite trees when all branches are explored
Optimality	Finds the shallowest goal in an unweighted tree	Does not generally guarantee the shallowest goal
Memory usage	Usually higher for wide trees	Often lower for wide, deep trees

5. Conclusion

Breadth-First Search and Depth-First Search are important search

techniques in Artificial Intelligence. Both algorithms begin at the root node A, but they explore the given state space tree differently.

BFS explores the tree level by level using a queue, while DFS explores one branch deeply before backtracking, using a stack or recursion.

For the given tree, the BFS traversal is:

A → B → C → D → E → D → G → C → F → B → F → G → G → H → E → G → H

The DFS traversal is:

A → B → D → C → G → F → G → H → E → C → D → B → E → F → G → H → G

Thus, BFS is suitable when the goal is to find the shallowest solution, whereas DFS is suitable when deep exploration of the state space is required.